α_ν = viscosity parameter → in the disk_prop folder you find $\alpha = 1e^{-3} = 10^{-3}$ and $\alpha = 1e^{-4} = 10^{-4}$

Appendix C: Comparison of disk sizes to analytic predictions of tidal truncation

Here we describe the analytic solutions to the models of tidal truncation and we then show the comparison of the observed dust disk radii to separation ratio with the analytic solutions.

C.1. Theoretical models of tidal truncations

Artymowicz & Lubow (1994) have described tidal truncation in circumstellar and circumbinary disks both analytically (in terms of resonant tidal interaction) and numerically. The location of disk truncation is set by balancing the resonant torques with the disk viscous torques and hence depends on the mass ratio, the orbital eccentricity, and the Reynolds number ($\mathcal{R} = (\alpha_\nu)^{-1}(r/H)^2$) in the disk. In the case of zero eccentricity, disk truncation is actually due mostly to nonresonant interaction (Papaloizou & Pringle 1977), in which case the truncation radius does not depend on viscosity and is just a function of the mass ratio, that we can express in terms of the mass parameter $\mu = M_2/(M_1 + M_2)$.

It is useful for the analysis carried out in Sect. 5.2 to obtain an analytical function $R_t(M_1, M_2, e, a)$, which for a given set of binary parameters returns the value of the truncation radius of the circumstellar disks predicted by the theory of Artymowicz & Lubow (1994).

Following the approach of Pichardo et al. (2005) we fit the results in both e and μ with an exponential function multiplied by the Roche Lobe radius ($R_{i,\mathrm{Egg}}$) of the appropriate star. The fitting function will therefore be

$$R_t(M_1, M_2, e, a) = R_{i,\mathrm{Egg}} \cdot (b \cdot e^c + h \cdot \mu^k), \quad \text{(C.1)}$$

where b, c, h, and k are the fitting parameters and

$$\frac{R_{i,\mathrm{Egg}}}{a} = \frac{0.49 \cdot q_i^{2/3}}{0.6 \cdot q_i^{2/3} + \ln(1 + q_i^{1/3})}, \quad \text{(C.2)}$$

where $q_1 = M_1/M_2$ and $q_2 = M_2/M_1$.

We note that Eq. (C.1) is composed of two terms. The first contains all the dependence on the eccentricity e ($R_{i,\mathrm{Egg}} b \cdot e^c$); the second ($R_{i,\mathrm{Egg}} h \cdot \mu^k$) describes how the zero-eccentricity truncation radius varies with varying (M_1, M_2, a). As explained above, the truncation in zero-eccentricity binaries is determined by the Papaloizou & Pringle (1977) mechanism: we can therefore obtain the value of h and k simply by fitting the results obtained by them. The fitting function (line) is overplotted to the data from Papaloizou & Pringle (1977; dots) in Fig. C.1, where the fitted parameters are $h = 0.88$ and $k = 0.01$. The exponent of μ is very small, the dependence on the masses is only inside $R_{i,\mathrm{Egg}}$, and in general the truncation occurs at 0.85–0.9 times the size of the Roche Lobe.

Now we calculate b and c by fitting the numerical results from Artymowicz & Lubow (1994) with

$$R_t(M_1, M_2, e, a) = R_{i,\mathrm{Egg}}(b \times e^c + 0.88 \cdot \mu^{0.01}). \quad \text{(C.3)}$$

Table C.1 reports the fitting parameters for some mass ratios and Reynolds numbers for both the circumprimary and the circumsecondary disk. The fitting parameters do not depend much on μ. For a general choice of μ we simply interpolate the fitting parameters reported in Table C.1.

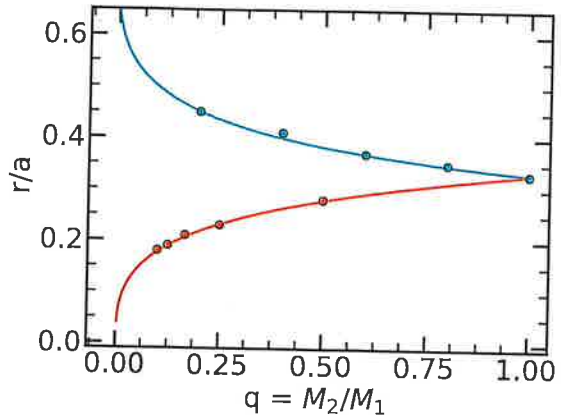


Fig. C.1. Tidal truncation radius predictions for a zero-eccentricity binary as a function of the mass ratio q. The solid lines represent the fitting function $R_{i,\mathrm{Egg}} 0.88 \cdot \mu^{0.01}$. In particular the blue line refers to the circumprimary disk, and the red line to the circumsecondary. Similarly, the blue dots are the theoretical predictions for the circumprimary and the red dots for the circumsecondary (data from Papaloizou & Pringle 1977). The yellow square refers to the equal mass case.

Table C.1. Best fit parameters for Eq. C.3 for different values of μ and $\mathcal{R}$, both for circumprimary and circumsecondary disks.

Circumprimary			Circumsecondary		
$\mathcal{R}$	b	c	$\mathcal{R}$	b	c
		$\mu = 0.1$			
10^4	−0.66	0.84	10^4	−0.81	0.98
10^5	−0.75	0.68	10^5	−0.81	0.80
10^6	−0.78	0.56	10^6	−0.83	0.69
		$\mu = 0.2$			
10^4	−0.72	0.88	10^4	−0.81	0.99
10^5	−0.78	0.72	10^5	−0.82	0.82
10^6	−0.80	0.60	10^6	−0.83	0.70
		$\mu = 0.3$			
10^4	−0.76	0.92	10^4	−0.79	0.97
10^5	−0.80	0.75	10^5	−0.82	0.81
10^6	−0.81	0.63	10^6	−0.83	0.69
		$\mu = 0.4$			
10^4	−0.77	0.95	10^4	−0.80	0.98
10^5	−0.81	0.78	10^5	−0.82	0.80
10^6	−0.82	0.66	10^6	−0.83	0.68
		$\mu = 0.5$			
10^4	−0.78	0.94	10^4	−0.79	0.95
10^5	−0.81	0.78	10^5	−0.81	0.78
10^6	−0.82	0.66	10^6	−0.82	0.66

C.2. Comparison with observations

We report here the plots of the comparison between the measured ratio of the dust disk radii and the projected separation with the expectations from analytic models, as described in Sect. 5.2.